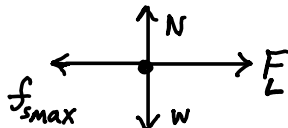


Name: Solution

These questions assess your understanding of the material from Chapters 4, 5, 6 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work.

*full credit is awarded only for complete work.
i.e. no points are awarded if complete work is not shown.*

1. Linda pushes a crate made of bronzebranch oak on a floor made of aetherium. The crate is initially at rest on the horizontal floor, until Linda pushes horizontally on the crate with a force of 330 N, the minimum force necessary to start its motion. Find the mass of the crate. Note that for bronzebranch oak on aetherium, $\mu_s = 0.95$ and $\mu_k = 0.77$.



$$\begin{aligned}\sum F_y &= ma_y = 0 \quad \because a_y = 0 \\ \vec{N} + \vec{w} &= 0 \\ N + (-w) &= 0 \\ N &= w \\ N &= mg\end{aligned}$$

ANSWER: 35 Kg

Linda pushes just strongly enough to overcome the friction force

$$\therefore F_L = f_{smax}$$

$$F_L = \mu_s N$$

$$F_L = \mu_s mg$$

$$m = \frac{F_L}{\mu_s g}$$

$$m = \frac{330 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{(0.95)(9.8 \frac{\text{m}}{\text{s}^2})} = \underline{35 \text{ kg}}$$

2. Caleb drives his 1160-kg car around a 210-m radius unbanked curve at 36.8 m/s (about 82 mph). Caleb's mass is 64 kg. Calculate the centripetal force exerted on the car as Caleb drives around the curve.

ANSWER: 7.89 kN

$$a_c = \frac{v^2}{r}$$

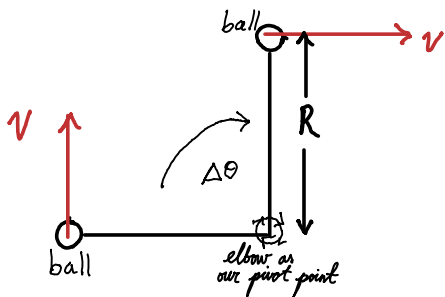
$$F_c = ma_c$$

$$F_c = \frac{mv^2}{r}$$

$$F_c = \frac{(1160 \text{ kg} + 64 \text{ kg})(36.8 \frac{\text{m}}{\text{s}})^2}{210 \text{ m}} = 7893 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} = 7893 \text{ N} = \underline{7.89 \text{ kN}}$$

3. While pitching in a baseball game, Jeremy brings his arm forward during a pitch, rotating his forearm about his elbow. If the velocity of the ball in his hand is 21.3 m/s and the ball is 0.33 m from the elbow joint, what is the angular velocity of the pitcher's forearm?

*find ω — this letter is omega.
it represents angular velocity.*

ANSWER: 64.54 rad/s

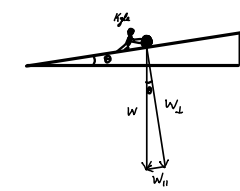
$$v = \omega R$$

$$\omega = \frac{v}{R}$$

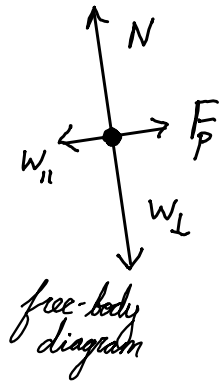
$$\omega = \frac{21.3 \frac{\text{m}}{\text{s}}}{0.33 \text{ m}} = 64.54 \frac{1}{\text{s}} = \underline{64.54 \frac{\text{rad}}{\text{s}}}$$

4. Kyle is trying to keep a 126.1-kg wagon at rest on a frictionless inclined plane. The plane makes an angle of 24.4° with the horizontal. What is the magnitude of the force Kyle must apply on the wagon to keep it from sliding down the plane?

ANSWER: 510.5 N



Sketch



free-body diagram

$$\Sigma F_{\parallel} = ma_{\parallel}$$

$$\vec{F}_p + \vec{W}_{\parallel} = 0 \because a_{\parallel} = 0$$

$$F_p + (-W_{\parallel}) = 0$$

$$F_p = W_{\parallel}$$

$$F_p = W \sin \theta$$

$$F_p = mg \sin \theta$$

$$F_p = (126.1 \text{ kg})(9.8 \text{ m/s}^2) \sin 24.4^\circ$$

$$F_p = 510.5 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} = \underline{510.5 \text{ N}}$$

5. As James prepares breakfast, he decides to conduct a quick experiment which reveals that the friction force between his steel spatula and his Teflon-coated frying pan is only 1.63 N as he moves the spatula. As he moves the spatula across the surface of the pan, what is the normal force that is exerted on his spatula? Note that for steel on Teflon, $\mu_s = 0.04$ and $\mu_k = 0.04$.

moving $\therefore$ use f_k not f_s

ANSWER: 40.75 N

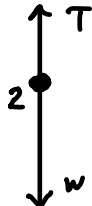
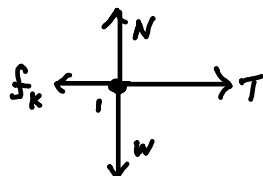
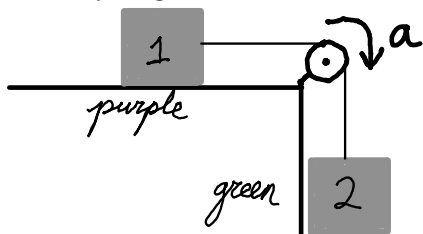
$$f_k = \mu_k N$$

$$N = \frac{f_k}{\mu_k}$$

$$N = \frac{1.63 \text{ N}}{0.04} = \underline{40.75 \text{ N}}$$

6. Ryan places a 2.9-kg purple wooden block on a wooden table. He then attaches a pulley to the table, threading a rope through the pulley to a 4.7-kg green block. This green block is hanging in the air by the rope and does not make contact with anything. What is the acceleration of the green hanging block? Note that for wood on wood, $\mu_s = 0.5$ and $\mu_k = 0.3$.

ANSWER: 4.939 m/s²



T: tension force, equal from 1 & 2

Purple block: $\Sigma F_x = ma$

$$T - f_k = ma$$

$$T - \mu_k N = ma$$

$$T - \mu_k mg = ma$$

$$T = m(a + \mu_k g)$$

$$\Sigma F_y = 0 \because a_y = 0$$

$$N - W = 0$$

$$N = W$$

$$N = mg$$

Green block: $\Sigma F_x = 0 \because a_x = 0$

$$\Sigma \vec{F}_y = m \vec{a} = m(-a) = -m_2 a$$

$$\vec{T} + \vec{W} = -m_2 a$$

$$T + (-W) = -m_2 a$$

$$T - m_2 g = -m_2 a$$

$$T = m_2 (g - a)$$

Equate the two equations we found for tension force & solve for a

$$m_1 (a + \mu_k g) = m_2 (g - a)$$

$$\frac{m_1}{m_2} a + \frac{m_1}{m_2} \mu_k g = g - a$$

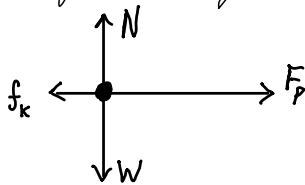
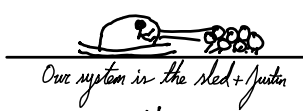
$$a \left(\frac{m_1}{m_2} + 1 \right) = g \left(1 - \frac{m_1}{m_2} \mu_k \right)$$

$$a = g \left(\frac{m_2 - m_1 \mu_k}{m_1 + m_2} \right)$$

$$a = 9.8 \frac{\text{m}}{\text{s}^2} \left(\frac{4.7 \text{ kg} - 2.9 \text{ kg} (0.3)}{2.9 \text{ kg} + 4.7 \text{ kg}} \right) = \underline{4.939 \frac{\text{m}}{\text{s}^2}}$$

7. A team of seven huskies pulls Justin and his sled across snow at 1.4 m/s. Justin and his loaded sled have a combined mass of 310 kg and the average mass of each dog is 26.1 kg. If each dog can pull with a force of 180 N, what is the magnitude of the acceleration of the sled? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.

ANSWER: 3.084 m/s^2



$$\begin{aligned}\sum \vec{F}_y &= m\vec{a}_y = 0 \therefore a_y = 0 \\ \vec{N} + \vec{W} &= 0 \\ N + (-W) &= 0 \\ N &= W = mg\end{aligned}$$

$$(F_p = 7F_h)$$

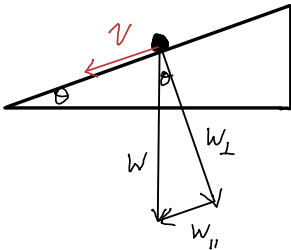
$$\begin{aligned}\sum \vec{F}_x &= m\vec{a}_x \\ \vec{F}_p + \vec{f}_k &= m\vec{a}_x \\ 7F_h + (-f_k) &= ma_x \\ 7F_h - \mu_k N &= ma_x \\ 7F_h - \mu_k mg &= ma_x \\ a_x &= \frac{7F_h}{m} - \mu_k g\end{aligned}$$

$$a_x = \frac{7(180\text{N})}{310\text{kg}} - (0.1)(9.8\text{m/s}^2) = 3.084 \frac{\text{m}}{\text{s}^2}$$

(Recall that $N = \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \therefore \frac{N}{\text{kg}} = \frac{\text{m}}{\text{s}^2}$)

8. Paul is skiing down a 21.5° slope. What is his acceleration? Note that for waxed wood on snow, $\mu_s = 0.14$ and $\mu_k = 0.1$.

ANSWER: 2.68 m/s^2



$$\begin{aligned}\sum \vec{F}_y &= 0 \therefore a_y = 0 \\ \vec{N} + \vec{W}_\perp &= 0 \\ N + (-W_\perp) &= 0 \\ N &= W_\perp = W \cos \theta \\ N &= Mg \cos \theta\end{aligned}$$

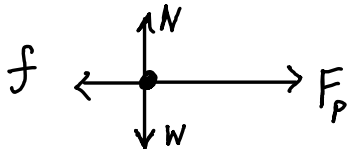
$$\begin{aligned}\sum \vec{F}_x &= m\vec{a}_x \\ \vec{W}_\parallel + \vec{f}_k &= m\vec{a}_x \\ W_\parallel - f_k &= ma \\ Mg \sin \theta - \mu_k N &= Ma \\ Mg \sin \theta - \mu_k Mg \cos \theta &= Ma \\ \vec{a} &= g(\sin \theta - \mu_k \cos \theta) \\ \vec{a} &= (9.8 \frac{\text{m}}{\text{s}^2})(\sin 21.5^\circ - 0.1 \cos 21.5^\circ) = 2.68 \frac{\text{m}}{\text{s}^2}\end{aligned}$$

9. Alexis is riding a motorcycle that can produce an acceleration of 4.4 m/s^2 while travelling at 80 km/h . The magnitude of the two forces resisting motion, friction and air resistance, total 510 N . The mass of Alexis and the motorcycle is 410 kg . What is the magnitude of the force that the motorcycle's rear tire exerts on the ground?

ANSWER: 2314 N

By Newton's 3rd Law, this is equal to the force that the ground exerts on the tire

motorcycle + Alexis is our system

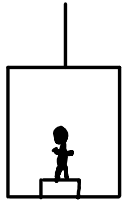


$$\begin{aligned}\sum \vec{F}_x &= m\vec{a}_x \\ \vec{F}_p + \vec{f} &= m\vec{a}_x \\ F_p - f &= ma_x \\ F_p &= ma_x + f\end{aligned}$$

$$F_p = (410\text{kg})(4.4\text{m/s}^2) + 510\text{N} = 2314 \text{ N}$$

$$\begin{aligned}\sum \vec{F}_y &= 0 \therefore a_y = 0 \\ \vec{N} + \vec{W} &= 0 \\ N - W &= 0 \\ N - mg &= 0 \\ N &= mg\end{aligned}$$

10. James, who has a mass of 61.1 kg, steps onto a scale that rests on the floor of an elevator. If the elevator accelerates downward at a rate of 8.8 m/s^2 , what will the scale read?



Scales report the normal force they exert on an object

ANSWER: 61.1 N

*let our system be James.
He accelerates downward,
 $\therefore F_{\text{net}}$ must be downward.*



$$\vec{F}_{\text{net}} = \sum \vec{F}_y = m \vec{a}_y$$

$$\vec{N} + \vec{w} = m \vec{a}_y$$

$$N - w = m(-a_y)$$

$$N - mg = -ma_y$$

$$N = m(g - a_y)$$

$$N = 61.1 \text{ kg} (9.8 \frac{\text{m}}{\text{s}^2} - 8.8 \frac{\text{m}}{\text{s}^2})$$

$$\underline{N = 61.1 \text{ N}}$$

11. Given that the planet Aurora orbits the planet Xanadu once every 107.0 days and that it is an average distance of 195,000 km from the center of Xanadu, calculate the period of an artificial satellite orbiting Xanadu at an average altitude of 1,680 km above Xanadu's surface. Note that the radius of Xanadu is 15,600 km.

ANSWER: 2.823 days or 67.74 hours



$$\frac{T_s^2}{T_A^2} = \frac{r_s^3}{r_A^3}$$

$r_s = 15600 \text{ km} + 1680 \text{ km}$
$r_s = 17280 \text{ km}$
$T_s = ?$
$r_A = 195000 \text{ km}$
$T_A = 107 \text{ days}$

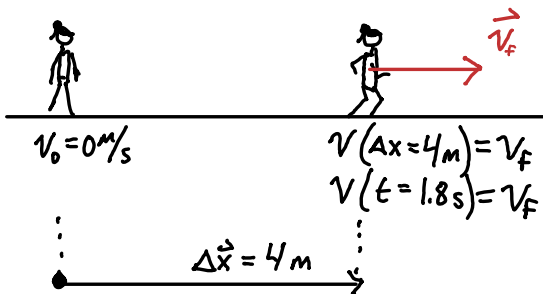
$$T_s = T_A \sqrt{\left(\frac{r_s}{r_A}\right)^3}$$

$$T_s = 107 \text{ days} \sqrt{\left(\frac{17280 \text{ km}}{195000 \text{ km}}\right)^3} = \underline{2.823 \text{ days}}$$

12. Chloe, a 76.5-kg sprinter, starts a 800-m race from rest and reaches the 4-m mark in 1.8 s. What is her average net force that she exerts on the ground, propelling herself forward?

ANSWER: 188.9 N

By Newton's 3rd Law, this is equal to the force that the ground exerts on Chloe



What is Chloe's acceleration during this motion?

$$\Delta x = v_0 t + \frac{1}{2} a_x t^2$$

$$a_x = \frac{2\Delta x}{t^2} \quad (\because v_{0x} = 0 \frac{\text{m}}{\text{s}})$$

Find the force acting on her

$$F_{\text{net},x} = m a_x$$

$$F_{\text{net},x} = m \left(\frac{2\Delta x}{t^2} \right)$$

$$F_{\text{net},x} = (76.5 \text{ kg}) \left(\frac{2(4 \text{ m})}{(1.8 \text{ s})^2} \right) = 188.9 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$\underline{F_{\text{net},x} = 188.9 \text{ N}}$$